

OPERATING SYSTEMS

UE25CS242A

Computer-System Architecture, OS Structure and Operations

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Acknowledgement, Prof. Pavan A C

Slides Credits for all the PPTs of this course

- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources:



1. "Operating System Concepts", Abraham Silberschatz, Peter Baer Galvin, Greg Gagne 10th Edition, John Wiley & Sons, Global Edition, 2023.
2. "Advanced Programming in the Unix Environment", Richard Stevens and Stephen A Rago, Pearson, 3rd edition, 2017.
3. Some conceptual text and diagram from "Operating Systems, Internals and Design Principles", William Stallings, 9th Edition, Pearson, 2018

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OPERATING SYSTEMS

Computer-System Architecture



- Most systems use a single general-purpose processor
  - Most systems have other special-purpose processors as well.
  - Device specific processors like disk, keyboard, graphic controller
  - Special-purpose processors run a limited number of instructions
  - Special-purpose processors are low-level components built into the hardware Managed by OS.
  - OS monitors the status.
- Example Disk controller microprocessor
  - Receives sequence of requests from CPU.
  - Implements its own disk queue and scheduling algorithm
  - Relieves the main CPU of the overhead of disk scheduling.

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Computer-System Architecture

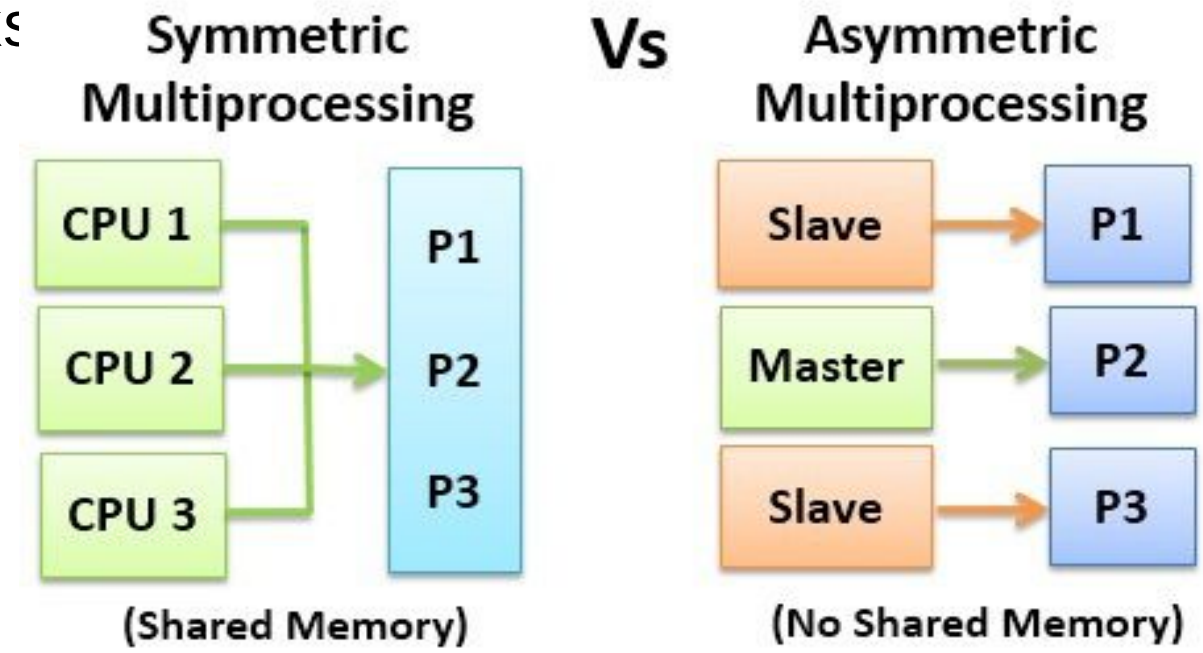


- **Multiprocessors** systems growing in use and importance
  - Also known as **parallel systems**, **tightly-coupled systems**
  - Advantages include:
    - **Increased throughput**
    - **Economy of scale**
    - **Increased reliability** – graceful degradation or fault tolerance

Multiprocessors

Two types of Multiprocessor Systems

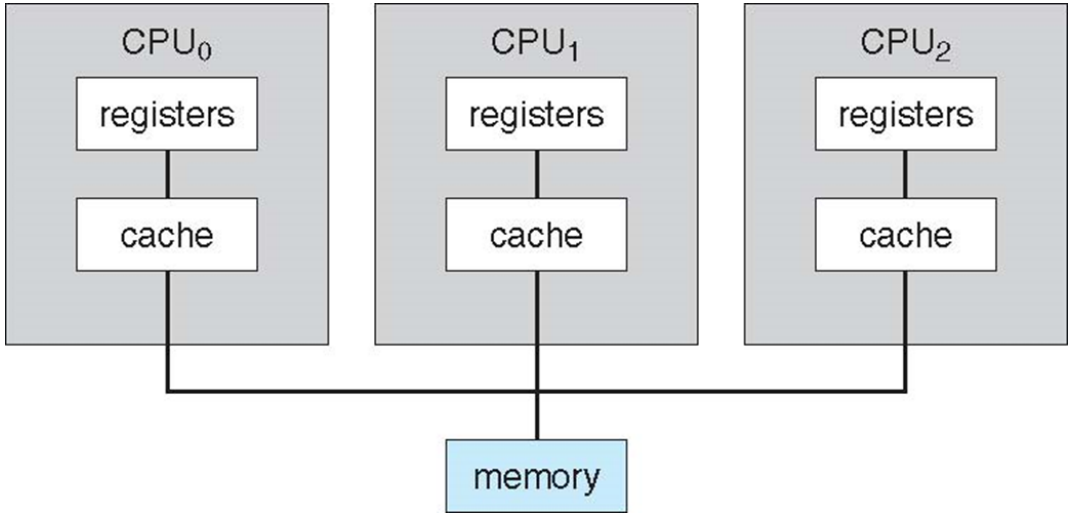
1. **Asymmetric Multiprocessing** – each processor is assigned a specific task.
2. **Symmetric Multiprocessing** – each processor performs all tasks



Symmetric Multiprocessing Architecture

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- In SMP all processors are peers; no boss–worker relationship exists between processors.
- Each processor has its own set of registers, as well as a private or local cache.
- All processors share physical memory.

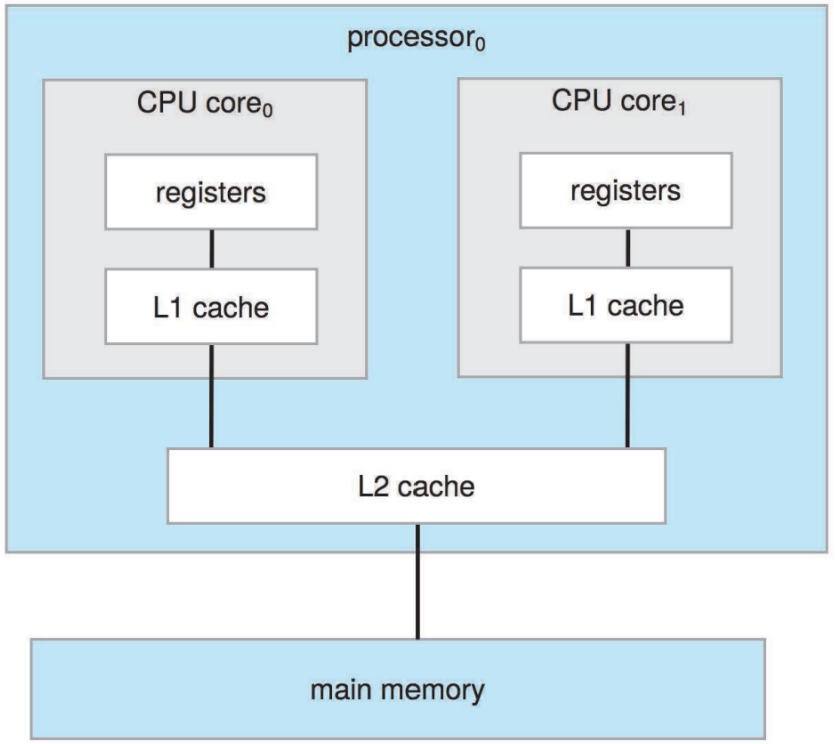


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Multi-Core Design

- A recent trend in CPU design is to include multiple computing cores on a single chip. Such multiprocessor systems are termed **multicore**.
- More efficient than multiple chips with single cores because on-chip communication is faster than between-chip communication.
- One chip with multiple cores uses significantly less power than multiple single-core chips.

Command to know the number of cores, cache details
$cat /proc/cpuinfo|more



A dual-core design with two cores placed on the same chip

Blade servers



- **Blade servers** are a recent development in which multiple processor boards, I/O boards, and networking boards are placed in the same chassis.
  - blade-processor board boots independently and runs its own operating system.
  - Some blade-server boards are multiprocessor as well, which blurs the lines between types of computers.
  - In essence, these servers consist of multiple independent multiprocessor systems.

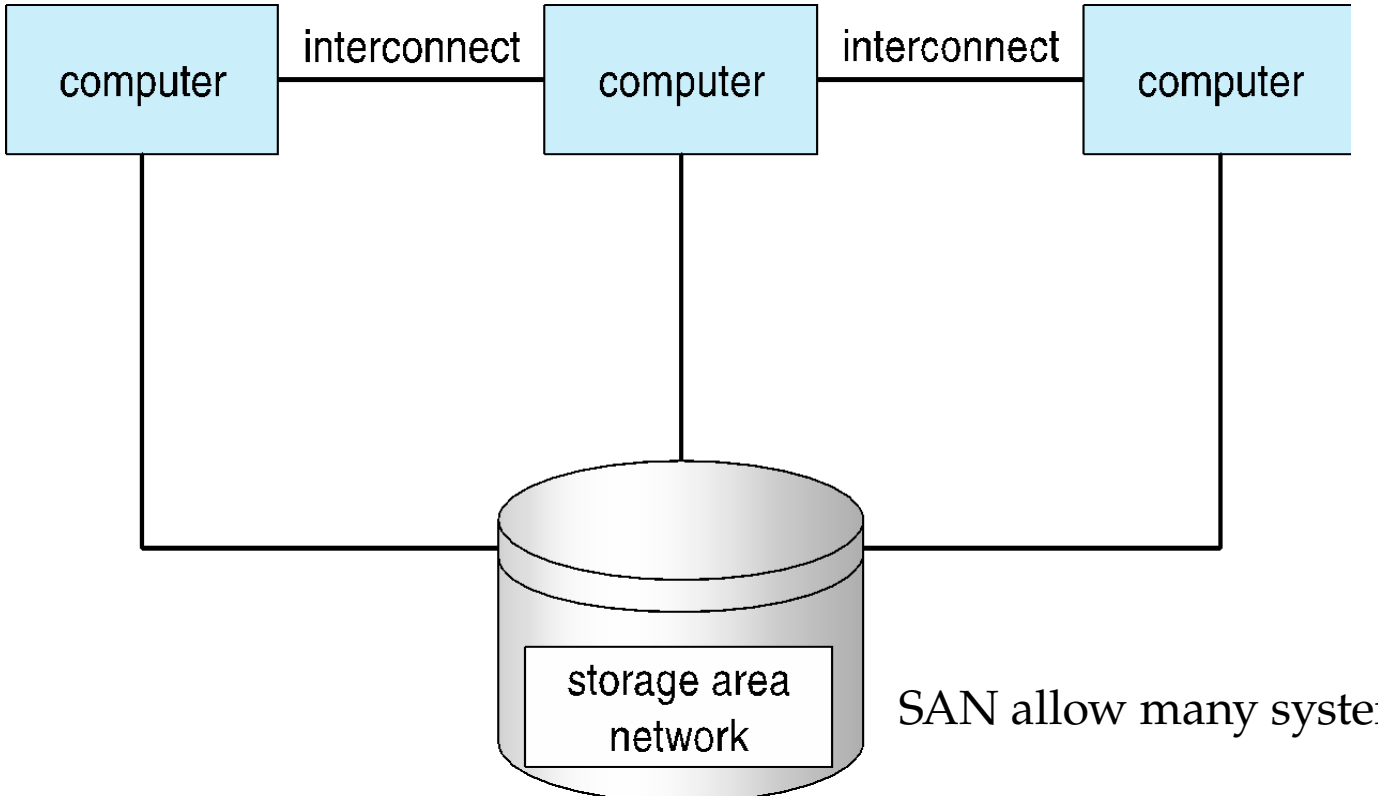
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Clustered Systems



- Like multiprocessor systems, but multiple systems working together
  - Usually sharing storage via a **storage-area network (SAN)**
  - Provides a **high-availability** service which survives failures
    - **Asymmetric clustering** has one machine in hot-standby mode
    - **Symmetric clustering** has multiple nodes running applications, monitoring each other
  - Some clusters are for **high-performance computing (HPC)**
    - Applications must be written to use **parallelization**
  - Some have **distributed lock manager** (**DLM**) to avoid conflicting operations (Ex: when multiple hosts access the same data on shared storage)

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Clustered Systems

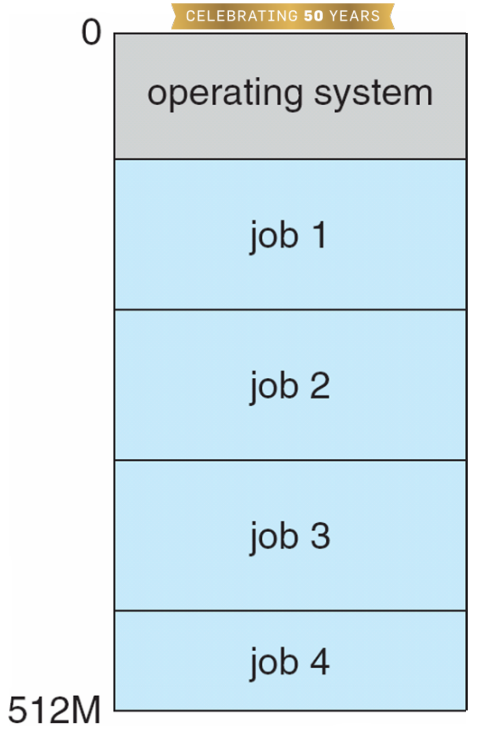


SAN allow many systems to attach to a pool of storage

General structure of a clustered system.

Operating-System Structure - Multiprogramming

- **Multiprogramming** (**Batch system**) needed for efficiency
  - Single user cannot keep CPU and I/O devices busy at all times
  - Multiprogramming organizes jobs (code and data) so CPU always has one to execute
  - A subset of total jobs in system is kept in memory
  - One job selected and run via **job scheduling**
  - When it has to wait (for I/O for example), OS switches to another job



Operating-System Structure - Multitasking



- **Timesharing** (**multitasking**) is logical extension in which CPU switches jobs so frequently that users can interact with each job while it is running, creating **interactive** computing
  - **Response time** should be < 1second
  - Each user has at least one program executing in memory ⇨**process**
  - If several jobs ready to run at the same time ⇨ **CPU scheduling**
  - If processes don't fit in memory, **swapping** moves them in and out to run

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Operating-System Operations



- **Interrupt driven** (hardware and software)
  - Hardware interrupt by one of the devices
  - Software interrupt (**exception** or **trap):**
    - Software error (e.g., division by zero)
    - Request for operating system service
    - Other process problems include infinite loop, processes modifying each other or the operating system

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Dual-Mode and Multimode Operation

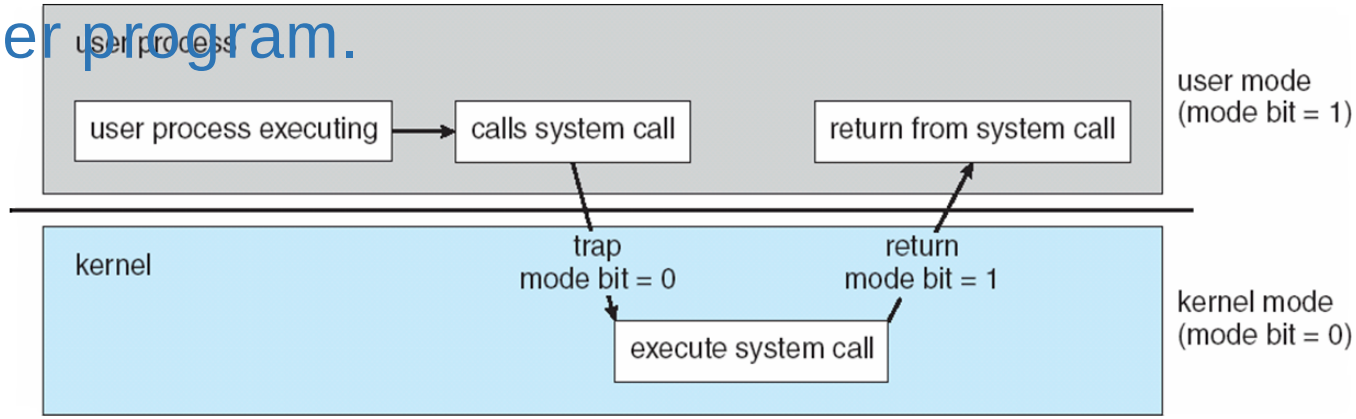


- **Dual-mode** operation allows OS to protect itself and other system components
  - **User mode** and **kernel mode**
  - **Mode bit** provided by hardware
    - Provides ability to distinguish when system is running user code or kernel code
    - Some instructions designated as **privileged**, only executable in kernel mode
    - System call changes mode to kernel, return from call resets it to user
- Increasingly CPUs support multi-mode operations
  - i.e. **virtual machine manager** (**VMM**) mode for guest **VMs**

Transition from user to kernel mode



- When a trap or interrupt occurs, hardware switches from user mode to kernel mode (changes the state of the mode bit to 0).
- When the request is fulfilled, the system always switches to user mode (by setting the mode bit to 1) before passing control to a user program.



Transition from user to kernel mode.

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Timer

- Timer to prevent infinite loop / process hogging resources
  - Timer is set to interrupt the computer after a specified period (fixed 1/60 sec or variable 1msec to 1sec)
  - A **variable timer** is generally implemented by a fixed-rate clock and a counter.
  - Operating system sets the counter (privileged instruction)
  - Every time the clock ticks, the counter is decremented.
  - When counter reaches zero, an interrupt occurs
  - Timer can be used to prevent a user program from running too long (terminate the program)



THANK YOU

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